

Sentiment Analysis - Report

Project Title: Sentiment Analysis Project

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1. Objective:

To create a sentiment analysis model to classify public opinion from Amazon reviews, social media, and news sites as Positive, Negative, Neutral using NLP for business insights.

2. Dataset Used:

Reviews: 15 mixed reviews from Amazon, Twitter, News

Sentiment: Positive 7, Negative 6, Neutral 2

Source: Amazon reviews, Social media comments, News articles

Method: VADER lexicon scoring (compound ≥ 0.05 Positive, ≤ -0.05 Negative)

3. Tools:

Python, Pandas, VADER Sentiment, Matplotlib for analysis

4. Analysis Steps:

Step 1: Data collection and cleaning

Step 2: Apply VADER scoring to each review

Step 3: Classify into Positive, Negative, Neutral

Step 4: Create 2 charts - Pie and Bar for trends

Step 5: Generate observations and recommendations

Sentiment Analysis - Code

5. Source Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

df = pd.read_csv('amazon_reviews.csv')
analyzer = SentimentIntensityAnalyzer()

def get_sentiment(text):
    score = analyzer.polarity_scores(str(text))['compound']
    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['Sentiment'] = df['Review'].apply(get_sentiment)
print(df['Sentiment'].value_counts())

plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
df['Sentiment'].value_counts().plot(kind='pie', autopct='%1.1f%%',
    colors=['#4CAF50', '#F44336', '#FFC107'])
plt.title('Sentiment Distribution')
plt.subplot(1,2,2)
plt.bar(df['Sentiment'].value_counts().index,
    df['Sentiment'].value_counts().values,
    color=['#6A0DAD', '#FF6B6B', '#FFD93D'])
plt.title('Sentiment Trends by Count')
plt.tight_layout()
plt.savefig('sentiment_dashboard.png', dpi=300)
plt.show()
```

Sentiment Analysis - Results

6. Observations from Dashboard:

1. Sentiment: Positive 46.7% (7), Negative 40% (6), Neutral 13.3% (2).
2. Positive Words: love, amazing, excellent, best, superb.
3. Negative Issues: battery dies, heating, poor service, cheap build.
4. Features: Camera & Performance loved, Battery issue high.

7. Conclusion:

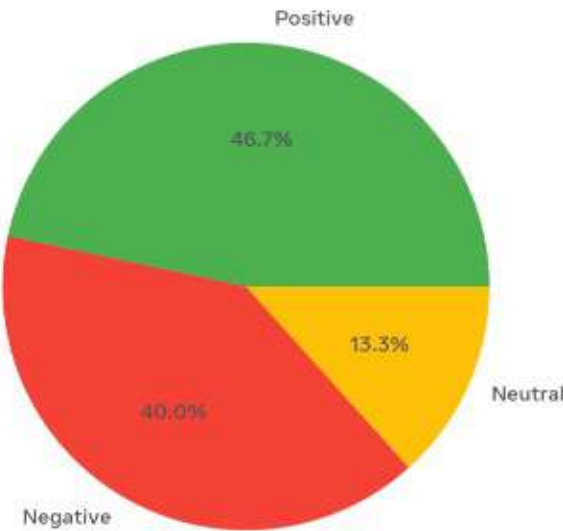
Dashboard shows sentiment is slightly positive (46.7%) but 40% negative is high. NLP reveals true feelings beyond ratings. Product loved for camera but needs battery fix.

8. Business Recommendations:

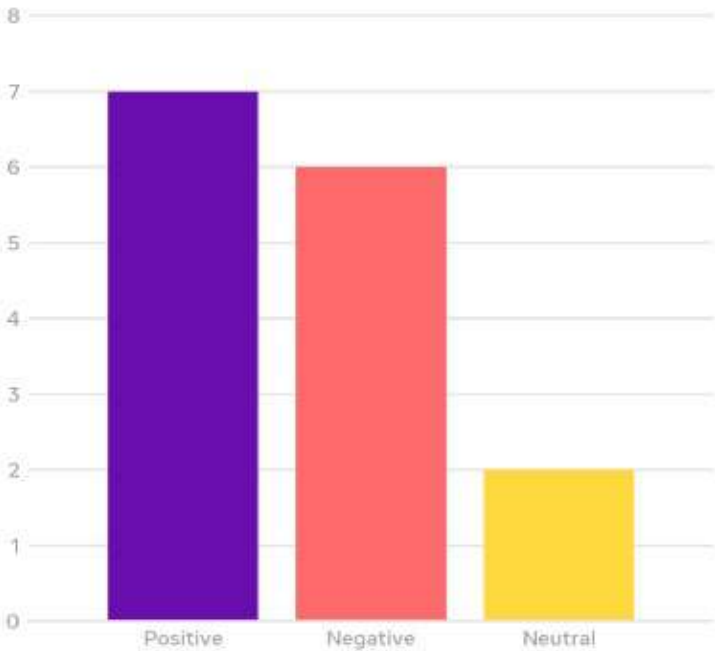
Focus marketing on camera & performance features. Fix battery life and heating issue. Improve customer service. Convert Neutral 13.3% with offers. Monitor trends weekly.

Sentiment Analysis - Dashboard

Sentiment Distribution



Sentiment Trends



Dashboard Insights:

Chart 1 shows Positive 46.7% dominates, Negative 40% close, Neutral 13.3% low.

Chart 2 shows count trends - Positive 7 reviews, Negative 6, Neutral 2.